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Inventor(s)	Noso; Terumitsu et al.

Sheet post-processing device and image forming system including same

Abstract

A sheet post-processing device includes a conveying member, a processing tray, a processing section, a reference plate, an alignment member, a holder drive unit, and a control unit. The alignment member includes a paddle holder supported in a swingable manner along a carry-in direction, and an alignment paddle supported in a rotatable manner at a swinging end of the paddle holder. The holder drive unit moves the alignment paddle in a reciprocating manner, between a reference position to retreat the paddle part above a top sheet placed on the processing tray and an acting position to allow the paddle part to contact with a top surface of the sheet. When the number of the sheets carried in onto the processing tray is increasing, the control unit continuously moves the acting position of the alignment paddle in a direction separating from the processing tray.

Inventors: Noso; Terumitsu (Osaka, JP), Egawa; Keisuke (Osaka, JP), Ueno; Yasunori (Osaka, JP), Shirasaki; Seiichi (Osaka, JP), Okada; Takehiko (Osaka, JP), Takano; Yusuke (Osaka, JP), Nishimura; Takuya (Osaka, JP)

Applicant: KYOCERA Document Solutions Inc. (Osaka, JP)

Family ID: 91762034

Assignee: KYOCERA Document Solutions Inc. (Osaka, JP)

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Primary Examiner: Bahls; Jennifer

Attorney, Agent or Firm: Stein IP, LLC

Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Applications No. 2023-002391 and 2023-054507 filed Jan. 11, 2023, and Mar. 30, 2023, respectively, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(2) The present disclosure relates to a sheet post-processing device that performs predetermined post-processing on a sheet after image formation by an image forming apparatus, and to an image forming system including the same.

(3) There is known a sheet post-processing device, which stacks a plurality of sheets after image formation by the image forming apparatus such as a copier or a printer, and performs a stapling process in which the stacked sheets are stapled as a whole by a stapler, a hole punching process in

which punch holes (perforation holes) are formed in the sheets by a hole punching device, or a folding process in which a fold line is formed in the sheets.

(4) This sheet post-processing device is equipped with a processing tray on which a predetermined number of sheets after image formation are placed. Then, the stapling process, a shift discharge process (sorting process), or the like is performed on the sheets placed on the processing tray. In addition, in order to smoothly perform the stapling process, the shift discharge process, or the like, the sheets on the processing tray are aligned using an alignment member such as an alignment paddle.

SUMMARY

(5) A sheet post-processing device according to one aspect of the present disclosure includes a conveying member, a processing tray, a processing section, a reference plate, an alignment member, a holder drive unit, and a control unit. The conveying member conveys a sheet. The processing tray stacks a plurality of sheets conveyed in a predetermined carry-in direction by the conveying member. The processing section performs predetermined post-processing on the sheets placed on the processing tray. The reference plate abuts an edge of the sheet carried in onto the processing tray on a downstream side in an alignment direction opposite to the carry-in direction, so as to align the sheet in the carry-in direction. The alignment member includes a paddle holder disposed above the processing tray so as to be supported in a swingable manner in the carry-in direction, and an alignment paddle having a paddle main body supported in a rotatable manner at a swinging end of the paddle holder and a paddle part protruding from the paddle main body in a radial direction, so as to move the sheet carried in onto the processing tray in the alignment direction for aiding alignment of the sheets. The holder drive unit swings the paddle holder, so as to move the alignment paddle in a reciprocating manner between a reference position to retreat the paddle part above a top sheet placed on the processing tray and an acting position to allow the paddle part to contact with the top surface of the sheet. The control unit controls the holder drive unit. When the number of the sheets carried in onto the processing tray is increasing, the control unit continuously moves the acting position of the alignment paddle in a direction separating from the processing tray.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating a structure of an image forming system constituted of a sheet post-processing device according to a first embodiment of the present disclosure, and an image forming apparatus to which the sheet post-processing device is connected.

(2) FIG. 2 is a cross-sectional side view schematically illustrating a structure of the sheet post-processing device of the first embodiment.

(3) FIG. 3 is a perspective view of a sheet stapling unit mounted on the sheet post-processing device of the first embodiment.

(4) FIG. 4 is a side view of the sheet stapling unit.

(5) FIG. 5 is a cross-sectional side view illustrating a processing tray, an alignment member, and their vicinity, in a state where an alignment paddle is positioned at a retreat position.

(6) FIG. 6 is a perspective view of the alignment paddle.

(7) FIG. 7 is a cross-sectional side view illustrating the processing tray, the alignment member, and their vicinity, in a state where the alignment paddle is positioned at an acting position.

(8) FIG. 8 is a flowchart illustrating an example of an adjustment control of an acting position of the alignment paddle in the sheet post-processing device of the first embodiment.

(9) FIG. 9 is a perspective view of the alignment member and its vicinity in the sheet post-processing device of a second embodiment of the present disclosure, viewed from a downstream

side in a discharge direction.

(10) FIG. **10** is a schematic diagram illustrating a relationship between a sheet deflection state and pressing forces of a first alignment paddle and a second alignment paddle in the sheet post-processing device of the second embodiment.

(11) FIG. **11** is a schematic diagram illustrating the sheet deflection state when the pressing force of the first alignment paddle is set smaller than that of the second alignment paddle, as a comparative example.

DETAILED DESCRIPTION

1. Structure of Image Forming System

(12) Hereinafter, with reference to the drawings, an embodiment of the present disclosure is described in detail. FIG. **1** is a schematic diagram illustrating a structure of an image forming system constituted of a sheet post-processing device **1** including a sheet stapling unit **92** according to a first embodiment of the present disclosure, and an image forming apparatus **200** to which the sheet post-processing device **1** is connected.

(13) As illustrated in FIG. **1**, the image forming apparatus **200** prints an image on a sheet (paper sheet), on the basis of image data input externally via a not-shown network communication unit, or image data read by an image reading unit **201** disposed at an upper part of the image forming apparatus **200**. In this embodiment, the image forming apparatus **200** is an inkjet recording device including recording heads (not shown) of respective colors, each of which has many nozzle holes for ejecting ink to a sheet.

(14) An operation panel **202** is disposed on a front of the image reading unit **201**. The operation panel **202** is an operation unit for receiving various setting inputs. For instance, a user can operate the operation panel **202** to input size information of the sheet. In addition, it is possible to operate the operation panel **202** to input the number of the sheet to be printed or to instruct start of a print job. A main body control unit **203** controls the whole operation of the image forming apparatus **200**, and controls individual units of the image forming apparatus **200**.

(15) The sheet post-processing device **1** is connected to a side face of the image forming apparatus **200** in an attachable and detachable manner. The sheet post-processing device **1** performs post-processing such as a hole punching process, a stapling process, or the like, on sheets after image formation (printing) by the image forming apparatus **200**. Note that the sheet post-processing device **1** is not limited to one that performs the post-processing on the sheets conveyed automatically from the image forming apparatus **200**, but may be one that conveys sheets set on a not-shown tray by a user to a position at which the post-processing can be performed, and performs the post-processing on the sheets.

2. Structure of Sheet Post-Processing Device

(16) FIG. **2** is a cross-sectional side view schematically illustrating a structure of the sheet post-processing device **1** of the first embodiment. As illustrated in FIG. **2**, the sheet post-processing device **1** includes a sheet carry-in entrance **2**, a first sheet conveying path **3**, a first sheet discharge unit **4**, a second sheet conveying path **5**, a second sheet discharge unit **6**, a third sheet conveying path **7**, a third sheet discharge unit **8**, a post-processing unit **9**, and a post-processing control unit (control unit) **10**.

(17) The sheet carry-in entrance **2** is an opening formed on the side face of the sheet post-processing device **1**, which faces the image forming apparatus **200**. The sheet conveyed from the image forming apparatus **200** to the sheet post-processing device **1** passes through the sheet carry-in entrance **2** and is carried into the sheet post-processing device **1**.

(18) The first sheet conveying path **3** extends in a substantially horizontal direction from the sheet carry-in entrance **2** to the first sheet discharge unit **4**, in a direction separating from the image forming apparatus **200** (to the left direction in FIG. **2**). Note that the direction from the sheet carry-in entrance **2** to the first sheet discharge unit **4** is referred to as a sheet conveying direction of the first sheet conveying path **3**. The sheet carry-in entrance **2** is positioned at an upstream end of the

first sheet conveying path **3** in the sheet conveying direction. The first sheet conveying path **3** includes a plurality of convey roller pairs **3r**, and conveys the sheet carried in from the sheet carry-in entrance **2** to the sheet post-processing device **1**, toward the downstream side in the sheet conveying direction.

(19) The first sheet discharge unit **4** is disposed at a side face of the sheet post-processing device **1**, which is opposite to the side face facing the image forming apparatus **200**. The first sheet discharge unit **4** is disposed at a downstream end of the first sheet conveying path **3** in the sheet conveying direction. The first sheet discharge unit **4** includes a first discharge outlet **41**, a first discharge roller pair **42**, and a first discharge tray **43**.

(20) The first discharge outlet **41** is positioned at the downstream end of the first sheet conveying path **3** in the sheet conveying direction. The first discharge roller pair **42** is disposed at the first discharge outlet **41**. The first discharge tray **43** is positioned on the downstream side of the first discharge outlet **41** in the sheet conveying direction. The sheet, which is conveyed along the first sheet conveying path **3** and reaches the first discharge outlet **41**, is discharged onto the first discharge tray **43** by the first discharge roller pair **42** after passing through the first discharge outlet **41**. The first discharge tray **43** is one of final discharge places of the sheets after post-processing by the sheet post-processing device **1**.

(21) The second sheet conveying path **5** branches from the first sheet conveying path **3** at a first junction (junction) **31** and extends laterally and upwardly to the second sheet discharge unit **6**, in the direction separating from the image forming apparatus **200** (to the left direction in FIG. **2**). The first junction **31** is disposed on the downstream side of a perforation unit **91** in the sheet conveying direction of the first sheet conveying path **3**. Note that the direction from the first junction **31** to the second sheet discharge unit **6** is referred to as a sheet conveying direction of the second sheet conveying path **5**. The first junction **31** is positioned at the upstream end of the second sheet conveying path **5** in the sheet conveying direction. The second sheet conveying path **5** includes a plurality of convey roller pairs **5r**, and allows the sheet conveyed along the first sheet conveying path **3** to branch at the first junction **31**, and conveys the same to the second sheet discharge unit **6**.

(22) The first junction **31** includes a first switching guide **311**. The first switching guide **311** rotates between a position for guiding the sheet, which is conveyed from the sheet carry-in entrance **2** along the first sheet conveying path **3**, to the first discharge outlet **41** along the first sheet conveying path **3**, and a position for guiding the sheet to the second sheet conveying path **5** after branching from the first sheet conveying path **3**. Furthermore, the first switching guide **311** rotates to a position for guiding the sheet, which passes through a later-described second folding conveying path **106** after a folding process, to the second sheet conveying path **5**. The first switching guide **311** is connected to a drive mechanism (not shown), and its operation is controlled by the post-processing control unit **10**.

(23) The second sheet discharge unit **6** is disposed above the first sheet discharge unit **4** on the side face of the sheet post-processing device **1**, which is opposite to the side face facing the image forming apparatus **200**. The second sheet discharge unit **6** is disposed at the downstream end of the second sheet conveying path **5** in the sheet conveying direction. The second sheet discharge unit **6** includes a second discharge outlet **61**, a second discharge roller pair **62**, and a second discharge tray **63**.

(24) The second discharge outlet **61** is positioned at the downstream end of the second sheet conveying path **5** in the sheet conveying direction. The second discharge roller pair **62** is disposed at the second discharge outlet **61**. The second discharge tray **63** is positioned on the downstream side of the second discharge outlet **61** in the sheet conveying direction. The sheet, which is conveyed along the second sheet conveying path **5** and reaches the second discharge outlet **61**, is discharged onto the second discharge tray **63** by the second discharge roller pair **62** after passing through the second discharge outlet **61**. The second discharge tray **63** is one of the final discharge places of the sheets after post-processing by the sheet post-processing device **1**. In addition, sheets

on which the post-processing is not performed, small size sheets, or the like are also discharged onto the second discharge tray **63**.

(25) The third sheet conveying path **7** branches from the first sheet conveying path **3** at a second junction **32**, and extends downward to the third sheet discharge unit **8**. Note that the direction from the second junction **32** to the third sheet discharge unit **8** is referred to as a sheet conveying direction of the third sheet conveying path **7**. The second junction **32** is positioned on the downstream side of the first junction **31** in the sheet conveying direction of the first sheet conveying path **3**, and is positioned at the upstream end of the third sheet conveying path **7** in the sheet conveying direction. The third sheet conveying path **7** includes a plurality of convey roller pairs **7r**, allows the sheet conveyed along the first sheet conveying path **3** to branch at the second junction **32**, and conveys the same to the third sheet discharge unit **8**.

(26) The second junction **32** includes a second switching guide **321**. The second switching guide **321** rotates between a position for guiding the sheet, which is conveyed from the sheet carry-in entrance **2** along the first sheet conveying path **3**, to the first discharge outlet **41** along the first sheet conveying path **3**, and a position for guiding the sheet, which is conveyed from the sheet carry-in entrance **2** along the first sheet conveying path **3** and is switched back after passing through the second junction **32**, to the third sheet conveying path **7**. The second switching guide **321** is connected to a drive mechanism (not shown), and its operation is controlled by the post-processing control unit **10**.

(27) The third sheet discharge unit **8** is disposed below the first sheet discharge unit **4** (near a lower end part of the sheet post-processing device **1**), on the side face of the sheet post-processing device **1**, which is opposite to the side face facing the image forming apparatus **200**. The third sheet discharge unit **8** includes a third discharge outlet **81**, a third discharge roller pair **82**, and a third discharge tray **83**.

(28) The third discharge outlet **81** is positioned at the downstream end of the third sheet conveying path **7** in the sheet conveying direction. The third discharge roller pair **82** is disposed at the third discharge outlet **81**. The third discharge tray **83** is positioned on the downstream side of the third discharge outlet **81** in the sheet conveying direction. The sheet, which is conveyed along the third sheet conveying path **7** and reaches the third discharge outlet **81**, is discharged onto the third discharge tray **83** by the third discharge roller pair **82** after passing through the third discharge outlet **81**. The third discharge tray **83** is one of the final discharge places of the sheets after post-processing by the sheet post-processing device **1**.

(29) The post-processing unit **9** performs predetermined post-processing on the sheets carried into the sheet post-processing device **1** after image formation by the image forming apparatus **200**. The post-processing unit **9** includes the perforation unit **91**, the sheet stapling unit **92**, a sheet folding unit **100**, and a binding unit **94**.

(30) The perforation unit **91** is disposed near the downstream side of the sheet carry-in entrance **2** of the first sheet conveying path **3**. The perforation unit **91** performs a perforation process on the sheet conveyed along the first sheet conveying path **3**, so as to form punch holes.

(31) The sheet stapling unit **92** is disposed near the upstream side of the first sheet discharge unit **4** in the sheet conveying direction of the first sheet conveying path **3**. The sheet stapling unit **92** performs the stapling process on a bundle of sheets formed by stacking a plurality of sheets, so as to staple the sheets. A detailed structure of the sheet stapling unit **92** will be described later.

(32) The sheet folding unit **100** is disposed on the downstream side of the perforation unit **91** and the upstream side of the stapling unit **92**, in the sheet conveying direction of the first sheet conveying path **3**. The sheet folding unit **100** performs the folding process on one sheet so as to form a fold line. The sheet folding unit **100** can perform the folding process on one sheet, such as half-fold, Z-fold, outer three-fold, inner three-fold, or the like.

(33) The binding unit **94** is disposed near the upstream side of the third sheet discharge unit **8** in the sheet conveying direction of the third sheet conveying path **7**. The binding unit **94** includes a

center-folding unit **941** and a center-stapling unit **942**. The binding unit **94** performs a center-folding process and a center-stapling process for folding and stapling the substantial center of a bundle of sheets formed by stacking a plurality of sheets, in the sheet conveying direction, so as to make a booklet.

(34) The post-processing control unit (control unit) **10** includes a CPU, a storage unit, and other electronic circuits and components (which are not shown). The post-processing control unit **10** is connected to the main body control unit **203** (see FIG. **1**) of the image forming apparatus **200** in a communicable manner. The post-processing control unit **10** receives a command from the main body control unit **203**, controls operations of individual components of the sheet post-processing device **1**, and performs processing related to functions of the sheet post-processing device **1**, using the CPU on the basis of control programs and data stored in the storage unit. The first sheet conveying path **3**, the first sheet discharge unit **4**, the second sheet conveying path **5**, the second sheet discharge unit **6**, the third sheet conveying path **7**, the third sheet discharge unit **8**, and the post-processing unit **9** each receive commands from the post-processing control unit **10** individually, and cooperate to perform post-processing on the sheet. Note that it may be possible that functions of the post-processing control unit **10** are performed by the main body control unit **203** of the image forming apparatus **200**.

3. Structure of Sheet Stapling Unit

(35) Next, a structure of the sheet stapling unit **92** is described. FIG. **3** is a perspective view of the sheet stapling unit **92** mounted on the sheet post-processing device **1**. FIG. **4** is a side view of the sheet stapling unit **92**.

(36) As illustrated in FIG. **3**, the sheet stapling unit **92** includes a processing tray **521**, a stapling unit **71**, and a reference plate **73**.

(37) The processing tray **521** is a rectangular tray that extends in a sheet width direction (arrow AA' direction) and in a carry-in direction. A plurality of sheets S (the bundle of sheets) to be stapled are placed on the processing tray **521**. In this case, the sheets S are carried in onto the processing tray **521** in the alignment direction toward the lower right direction in FIG. **4** (arrow B direction), which is the opposite direction to the carry-in direction. The bundle of sheets after the stapling process is eventually sent out by the first discharge roller pair **42** (see FIG. **2**) in the opposite direction to the above alignment direction (to the upper left direction in FIG. **4**), and is discharged onto the first discharge tray **43** (see FIG. **2**). The downstream side of the processing tray **521** in the carry-in direction (the lower left side in FIG. **3**) supports a lower discharge roller **421** constituting the first discharge roller pair **42**.

(38) The processing tray **521** includes a tray middle part **522** and width restricting members **523**. The tray middle part **522** is disposed on the top surface part of the processing tray **521** in the middle part in the sheet width direction. The tray middle part **522** is a sheet-like member fixed on the processing tray **521** with a slight height.

(39) The width restricting members **523** are disposed in pair on both sides of the tray middle part **522** in the sheet width direction. The width restricting members **523** restrict a position in the sheet width direction of the sheet S carried in onto the processing tray **521**. The width restricting member **523** is constituted of a sheet-like member similarly to the tray middle part **522**, and has a sidewall standing upward at an end in the sheet width direction. The processing tray **521** is provided with a guide groove **524** formed to extend in the sheet width direction. The width restricting members **523** can be moved in a reciprocating manner along the guide groove **524** in the sheet width direction, by a drive mechanism such as a rack and pinion gear (not shown). In this embodiment, every time when the sheet is carried in onto the processing tray **521**, the width restricting members **523** are driven by the drive mechanism to move in a reciprocating manner. As a result, the sheet placed on the processing tray **521** is aligned in the sheet width direction.

(40) The stapling unit **71** is disposed so as to face an end edge of the sheet on the front end side in the alignment direction (the right side in FIG. **4**). The stapling unit **71** can be moved by drive force

of a stapling unit drive motor M1 along the end edge of the sheet in the sheet width direction (arrow AA' direction) perpendicular to the carry-in direction, and performs the stapling process on the bundle of the sheets.

(41) As illustrated in FIG. 4, the stapling unit 71 includes a staple main body part 711 and a staple movable part 712. The staple main body part 711 is a main body part of the stapling unit 71, in which a plurality of staple needles (not shown) are housed. The staple movable part 712 can move in the up and down direction, so as to insert the staple needle into the sheets. Between the staple main body part 711 and the staple movable part 712, there is formed a recess 713 into which the end edge of the sheet enters.

(42) The reference plates 73 are fixed at three positions separated in the sheet width direction, so as to face the end of the processing tray 521 on the downstream side (the upper right side in FIG. 3 and the lower right side in FIG. 4) in the alignment direction. The reference plate 73 has a substantial U-shape that opens to the upstream side (the upper left side in FIG. 4) in the alignment direction, in a cross section perpendicular to the sheet width direction. The reference plate 73 abuts the end edge of the sheet carried in onto the processing tray 521, so as to align the sheet in the carry-in direction.

4. Structure of Processing Tray and its Vicinity of Sheet Stapling Unit

(43) FIG. 5 is a cross-sectional side view illustrating a structure of the processing tray 521, an alignment member 55, and their vicinity. As illustrated in FIG. 5, a carry-in roller pair 54 is disposed above the processing tray 521. The carry-in roller pair 54 is constituted of an upper carry-in roller 54a and a lower carry-in roller 54b.

(44) A sheet detection sensor 93 is disposed in a vicinity of the carry-in roller pair 54. The sheet detection sensor 93 detects timing when the sheet S passes the carry-in roller pair 54. As the sheet detection sensor 93, for example, a photo interrupter (PI) sensor is used, which includes a detection unit constituted of a light emitting part and a light receiving part.

(45) A tapping member 53 and the alignment member 55 are disposed on the downstream side of the carry-in roller pair 54 in the carry-in direction of the sheet S (the left side in FIG. 5). The tapping member 53 is supported in a swingable manner along the carry-in direction of the sheet S. The tapping member 53 swings downward at timing when the rear end of the sheet S passes the carry-in roller pair 54, so as to tap the sheet S downward to be along the processing tray 521.

(46) The alignment members 55 are disposed at a plurality of places (four places in this embodiment) in the sheet width direction (the direction perpendicular to the paper of FIG. 5). The alignment member 55 moves (switchbacks) the sheet S carried in onto the processing tray 521 in the alignment direction to approach the reference plate 73, so as to aid alignment of the sheet S. The alignment member 55 includes a paddle holder 56 and an alignment paddle 57.

(47) The paddle holder 56 is supported in a swingable manner along the carry-in direction of the sheet S above the processing tray 521. A rotation drive force from a holder drive motor M2 is input to a swing shaft 56a of the paddle holder 56. As the holder drive motor M2, a stepping motor is used, which can be accurately controlled in its rotation direction and rotation amount (rotation angle) by pulse control. In addition, there is disposed an HP detection sensor (not shown) for detecting a reference position (home position) of the paddle holder 56 (the alignment paddle 57). The post-processing control unit 10 estimates height of the alignment paddle 57 (distance from the processing tray 521), on the basis of a detection signal of the HP detection sensor and the number of the sheets S placed on the processing tray 521.

(48) FIG. 6 is a perspective view of the alignment paddle 57. The alignment paddle 57 includes a paddle main body 571 and paddle parts 572. The paddle main body 571 has a cylindrical shape, in which a fulcrum part 571a is formed. The fulcrum part 571a is a through hole penetrating along the center axis of the paddle main body 571, and is supported in a rotatable manner at the swinging end of the paddle holder 56 (the left end in FIG. 5). The paddle parts 572 are rubber sheets, which protrude in tangent directions (to the upstream side in the rotation direction) from symmetric

positions on an outer periphery of the paddle main body **571**, with respect to the center of the fulcrum part **571a**.

(49) A rotation drive force is input to the alignment paddle **57** from a drive source such as a motor (not shown), in a direction to send out the sheet **S** in the alignment direction (the counterclockwise direction in FIG. **5**). When the paddle part **572** of the alignment paddle **57** rotates in contact with the top surface of the sheet **S** carried in onto the processing tray **521**, the sheet **S** is moved in the alignment direction, and the end edge of the sheet abuts the reference plate **73** for alignment. The alignment paddle **57** rotates by a preset number of rotations (for example, by two rotations), and then stops.

(50) The swinging of the paddle holder **56** is controlled based on the detection timing of the sheet detection sensor **93**. Specifically, at timing when the sheet detection sensor **93** detects that the front end of the sheet **S** passes the carry-in roller pair **54**, the paddle holder **56** is swung in the upward direction. As a result, the paddle part **572** separates from the top surface of the processing tray **521** (or the sheet **S** placed on the processing tray **521**).

(51) A support member **58** is disposed below the processing tray **521**. The support member **58** is a rod-like member that has a predetermined width in the sheet width direction and extends in an arc shape in the discharge direction, and is disposed below the first discharge outlet **41**. More specifically, the support member **58** is disposed below the processing tray **521**, and below a discharge path of the sheet **S** that is discharged from the first discharge roller pair **42** along the processing tray **521**. In this embodiment, the support members **58** are disposed at two positions in the sheet width direction separated by predetermined distances from the middle part of the processing tray **521** in the sheet width direction.

(52) The support member **58** can move between a protruding position to protrude from the first discharge roller pair **42** to the downstream side in the discharge direction (the left side in FIG. **5**), which is the position illustrated by a solid line in FIG. **5**, and a retreat position to retreat from the first discharge roller pair **42** to the upstream side in the discharge direction (the right side in FIG. **5**), which is the position illustrated by a broken line in FIG. **5**. When the sheet **S** is carried in onto the processing tray **521** (when it is switched back), the support member **58** is positioned at the protruding position, so as to support a part of the sheet **S** protruding on the first discharge tray **43**.

(53) FIG. **5** illustrates a state just before a new sheet **S** is carried in onto the processing tray **521**, in which the paddle holder **56** is swung in the upward direction (clockwise direction), and the alignment paddle **57** is positioned at a position apart from the processing tray **521** (the reference position). In addition, a nip between the lower discharge roller **421** and an upper discharge roller **422** constituting the first discharge roller pair **42** is released. In this way, the sheet **S** carried in onto the processing tray **521** by the carry-in roller pair **54** temporarily passes the first discharge roller pair **42** and protrudes on the first discharge tray **43**.

(54) Further, at timing when the rear end of the new sheet **S** carried in onto the processing tray **521** passes the carry-in roller pair **54**, the paddle holder **56** is swung in the reverse direction (the counterclockwise direction) as illustrated in FIG. **7**. As a result, the alignment paddle **57** is positioned at a position where the paddle part **572** makes contact with the top surface of the sheet **S** (the acting position). The operation described above is repeatedly performed every time when the sheet **S** is carried in, and hence the paddle part **572** can securely make contact with the top surface of the sheet **S** while avoiding interference between the paddle part **572** and the front end of the sheet **S** carried in onto the processing tray **521**.

5. Adjustment Control of Acting Position of Alignment Paddle

(55) Next, adjustment control of the acting position of the alignment paddle **57** is described. In this embodiment, the acting position of the alignment paddle **57** is changed based on the number of the sheets **S** carried in onto the processing tray **521**. More specifically, when the number of the sheets **S** carried in onto the processing tray **521** is increasing, the acting position of the alignment paddle **57** is being separated from the processing tray **521**.

(56) Specifically, the post-processing control unit **10** counts the number of the sheets **S** carried in onto the processing tray **521**. Then, it sends a control signal to the holder drive motor **M2**, so as to move upward a stop position of the paddle holder **56** every time when one sheet **S** (or a predetermined number of sheets) is carried in. In this way, the acting position of the alignment paddle **57** is also moves upward continuously.

(57) According to the structure described above, in accordance with the upward movement of the position of the top surface of the sheet **S** due to increase in the number of the sheets **S** carried in onto the processing tray **521**, the acting position of the alignment paddle **57** also moves upward. As a result, regardless of the number of the sheets **S** that is carried in, the contact state of the paddle part **572** with the top surface of the sheet **S** can be maintained to be constant. Therefore, from a state where a small number of the sheets **S** are placed on the processing tray **521** to a state where the maximum possible number of the sheets **S** are placed on the same, the sheet **S** can be securely moved in the alignment direction, and the end edge of the sheet can abut the reference plate **73** for alignment.

(58) Here, the conveying power of the alignment paddle **57** necessary for alignment of the sheet **S** varies largely depending on a type of the sheet **S** (size, thickness, basis weight, grain direction, or surface smoothness). If the sheet **S** is a paper sheet, fibers of the sheet **S** tend to align along the direction in which the sheet flows in a paper machine. This direction in which fibers align is referred to as the “grain direction” of the sheet **S**.

(59) In addition, if the image forming apparatus **200** connected to the sheet post-processing device **1** is an inkjet recording device, a conveying load for pulling the sheet **S** onto the processing tray **521** varies depending also on the amount of ink used for image formation. Therefore, it may be difficult to securely convey the sheet **S** and to align the same only by setting the acting position based on the number of the carried-in sheets **S**.

(60) Therefore, in this embodiment, the acting position of the alignment paddle **57** is adjusted based on output information about characteristics of the sheet **S**. Specifically, the output information about characteristics of the sheet **S** (a type of the sheet **S** and the amount of ink) is input from the image forming apparatus **200**. The post-processing control unit **10** adjusts the acting position of the alignment paddle **57**, on the basis of the output information about characteristics of the sheet **S** that is input.

(61) FIG. **8** is a flowchart illustrating an example of an adjustment control of the acting position of the alignment paddle **57**, which is performed by the sheet stapling unit **92** of the first embodiment. With reference to FIGS. **1** to **7** as necessary, along the steps illustrated in FIG. **8**, adjusting procedure of the acting position of the alignment paddle **57** is described. Note that it is supposed that, in an initial state, the acting position of the alignment paddle **57** is set at a reference acting position according to reference output information (for example, the output information that is used most frequently). In addition, in the following description, the output information about characteristics of the sheet **S** is also referred to simply as output information of the sheet.

(62) When a stapling process command for the sheets **S** is input from the main body control unit **203** of the image forming apparatus **200**, the output information of the sheet **S** is input together with the stapling process command (Step **S1**). Among the output information of the sheet **S**, information about a type of the sheet **S** such as size, thickness, basis weight, grain direction, or surface smoothness of the sheet **S** is input from the operation panel **202** of the image forming apparatus **200**. For instance, it may be possible to store a manufacturer, a trade name, a product number, or the like of the sheet **S** in association with the corresponding information about a type of the sheet **S** in the main body control unit **203** in advance. In this way, only when the user selects a manufacturer, a trade name, a product number, or the like of the sheet **S** using the operation panel **202**, the main body control unit **203** can recognize the information about a type of the sheet **S** to be used.

(63) As for the information about the amount of ink, the main body control unit **203** calculates the

amount of ink to be used for image formation, on the basis of image data sent from a host device such as a personal computer or the image reading unit **201**, and sends out the same.

(64) The post-processing control unit **10** determines whether or not it is necessary to correct the acting position of the alignment paddle **57**, on the basis of the output information of the sheet **S** that is input (Step **S2**). If it is determined that it is necessary to correct the acting position (Yes in Step **S2**), the post-processing control unit **10** corrects the acting position of the alignment paddle **57** from the reference acting position (Step **S3**).

(65) For instance, if a size, a thickness, a basis weight, or a surface friction force of the sheet **S** increases, the conveying power necessary for moving the sheet **S** is increased. Specifically, if the sheet **S** is a paper sheet made in Europe, it has a tendency to have less surface smoothness and a larger friction force between sheets than a paper sheet made in Japan or USA. In addition, if the grain direction of the sheet **S** is parallel to the sheet width direction, the conveying power necessary for moving the sheet **S** is larger than in the case where the grain direction of the sheet **S** is parallel to the carry-in direction. Therefore, if a size, a thickness, a basis weight, or a surface friction force of the sheet **S** is larger than the reference output information, or if the grain direction of the sheet **S** is parallel to the sheet width direction, the acting position of the alignment paddle **57** is corrected to be closer to the processing tray **521** than the reference acting position.

(66) On the contrary, if a size, a thickness, a basis weight, or a surface friction force of the sheet **S** is smaller than the reference output information, or if the grain direction of the sheet **S** is parallel to the carry-in direction, the acting position of the alignment paddle **57** is corrected to be more distant from the processing tray **521** than the reference acting position. The correction amount from the reference acting position is determined based on a correction table, which defines a relationship between a size, a thickness, a basis weight, a grain direction, or surface smoothness of the sheet **S** and the acting position (drive pulses of the holder drive motor **M2**) corresponding to them.

(67) In addition, if the amount of ink used for image formation increases, weight of the sheet **S** is increased, and hence the conveying power necessary for moving the sheet **S** is increased. Therefore, if the amount of ink is more than the reference output information, the acting position of the alignment paddle **57** is corrected to be closer to the processing tray **521** than the reference acting position. On the contrary, if the amount of ink is less than the reference output information, the acting position of the alignment paddle **57** is corrected to be more distant from the processing tray **521** than the reference acting position. The correction amount from the reference acting position is determined based on a correction table, which defines a relationship between the amount of ink and the acting position (drive pulses of the holder drive motor **M2**) corresponding to it.

(68) If it is determined that it is not necessary to correct the acting position (No in Step **S2**), the post-processing control unit **10** proceeds to the next stop without changing the acting position of the alignment paddle **57** from the reference acting position.

(69) Next, the post-processing control unit **10** starts to carry in the sheet **S**, which is carried into the sheet post-processing device **1** through the sheet carry-in entrance **2**, onto the processing tray **521** (Step **S4**). Specifically, when the sheet detection sensor **93** detects that the rear end of the sheet **S** passes the carry-in roller pair **54**, the post-processing control unit **10** (see FIG. **2**) controls the tapping member **53** to tap the rear end of the sheet **S** to be along the processing tray **521**, and then controls the holder drive motor **M2** to move the paddle holder **56** downward by a predetermined amount. In this way, the alignment paddle **57** moves in the acting position, and the paddle part **572** makes contact with the top surface of the sheet **S**. In this state, the alignment paddle **57** is rotated, and hence the sheet **S** is pulled in along the processing tray **521** in the alignment direction (arrow **B** direction).

(70) After that, the sheet **S** is further sent to the downstream side in the alignment direction by the alignment paddle **57**, and is aligned by the width restricting member **523** (see FIG. **3**) in the sheet width direction, and is aligned by the reference plate **73** in the carry-in direction, so as to be stacked.

(71) The post-processing control unit **10** counts the number of sheets **S** carried in onto the processing tray **521** (Step **S5**). Further, every time when one sheet **S** (or a predetermined number of sheets **S**) is carried in, the stop position of the paddle holder **56** is moved upward, and the acting position of the alignment paddle **57** is continuously moved upward.

(72) Next, the post-processing control unit **10** determines whether or not a predetermined number of sheets **S** have been carried in onto the processing tray **521** (Step **S6**). If a predetermined number of sheets **S** have not carried in (No in Step **S6**), the process returns to Step **S4**, and the carrying in of the sheets **S** onto the processing tray **521** and the counting of the number of the carried-in sheets are continued.

(73) If a predetermined number of sheets **S** have been carried in (Yes in Step **S6**), the post-processing control unit **10** sends a control signal to the stapling unit drive motor **M1** (see FIG. **3**) so as to move the stapling unit **71** to a predetermined staple position. After the stapling unit **71** is moved to the predetermined staple position, the post-processing control unit **10** sends a control signal to the stapling unit **71**, so as to perform the stapling process on the plurality of sheets **S** aligned by the reference plate **73** (Step **S7**).

(74) Next, the post-processing control unit **10** controls the first discharge roller pair **42** to be the contact state (to form the nip), and controls the first discharge roller pair **42** to rotate in the discharge direction. In this way, the bundle of the sheets **S** after the stapling process is discharged by the first discharge roller pair **42** onto the first discharge tray **43**, which are illustrated in FIG. **2** (Step **S8**). After the bundle of the sheets **S** are discharged, the post-processing control unit **10** sends a control signal to the stapling unit drive motor **M1** so as to move the stapling unit **71** to a standby position (Step **S9**).

(75) After that, the post-processing control unit **10** determines whether or not the stapling process is finished (Step **S10**). If the stapling process is continuing (No in Step **S10**), the process returns to Step **S4**, so as to repeat carrying in of the sheets **S** onto the processing tray **521**, the counting of the number of the carried-in sheets, executing of the stapling process, discharge of the bundle of sheets, and the moving of the stapling unit **71** to the standby position (Steps **S4** to **S10**). If the stapling process is finished (Yes in Step **S10**), the process is finished as it is.

(76) According to the control example illustrated in FIG. **8**, the acting position of the alignment paddle **57** is corrected to be an optimal position, on the basis of the output information of the sheet **S**. Therefore, without being affected by a type of the sheet **S** or amount of ink after image formation by the inkjet recording device, the sheet **S** can be securely conveyed and aligned.

(77) In addition, as the acting position is continuously moved upward in accordance with the number of the sheets **S** carried in onto the processing tray **521**, the contact state of the paddle part **57** with the top surface of the sheet **S** can be maintained to be constant regardless of the number of the carried-in sheets **S**.

(78) FIG. **9** is a perspective view of the alignment member **55** and its vicinity in the sheet post-processing device **1** of a second embodiment of the present disclosure, viewed from the downstream side in the discharge direction. In this embodiment, the alignment paddle **57** of the alignment member **55** includes a pair of first alignment paddles **57a** disposed in the middle part in the sheet width direction, and a pair of second alignment paddles **57b** disposed outside of the first alignment paddles **57a** in the sheet width direction. The structures of the first alignment paddle **57a** and the second alignment paddle **57b** are the same as that of the alignment paddle **57** of the first embodiment illustrated in FIG. **6**.

(79) FIG. **10** is a schematic diagram illustrating a relationship between a deflection state of the sheet **S** and pressing forces of the first alignment paddle **57a** and the second alignment paddle **57b** in the sheet post-processing device **1** of the second embodiment. The white arrows in FIG. **10** indicate magnitudes of pressing forces **F1** and **F2** of the first alignment paddle **57a** and the second alignment paddle **57b** applied to the sheet **S**. As illustrated in FIG. **10**, the pressing force (first pressing force) **F1** of the first alignment paddle **57a** is set larger than the pressing force (second

pressing force) F2 of the second alignment paddle 57b. In other words, the pulling force acting on the sheet S is larger on the inside than the outside in the width direction of the sheet S.

(80) In this way, the pulling timing of the sheet S by the second alignment paddle 57b on the outside in the width direction is a little delayed from the pulling timing of the sheet S by the first alignment paddle 57a on the inside in the width direction. As a result, a force is applied to the sheet S in the direction to expand it outside in the width direction, and hence the sheet is pulled in onto the processing tray 521 without a deflection in the width direction.

(81) FIG. 11 is a schematic diagram illustrating the deflection state of the sheet S when the pulling force of the first alignment paddle 57a is set smaller than that of the second alignment paddle 57b, as a comparative example. As illustrated in FIG. 11, if the pressing force F1 of the first alignment paddle 57a is set smaller than the pressing force F2 of the second alignment paddle 57b, the pulling force acting on the sheet S becomes larger on the outside than the inside of the sheet S in the width direction. As a result, the pulling timing of the sheet S on the outside in the width direction becomes earlier, and hence a force is applied to the sheet S in the direction to pull the sheet S to the inside in the width direction.

(82) Therefore, the sheet S tends to be deflected in the width direction, so that a wrinkle may occur in the sheet S or resilience of the deflected sheet S may cause a rebound. Note that even if the pressing forces of the first alignment paddle 57a and the second alignment paddle 57b are set equal to each other, a variation of the pressing force may cause a deflection of the sheet S in the width direction.

(83) As illustrated in FIG. 10, by setting the pressing force F1 of the first alignment paddle 57a larger than the pressing force F2 of the second alignment paddle 57b in advance, even if there is a variation in the pressing force, the deflection of the sheet in the width direction, and wrinkle or rebound of the sheet S due to the same can be suppressed. Further, alignment property of the sheet S pulled in onto the processing tray 521 can be improved.

(84) The pressing forces of the first alignment paddle 57a and the second alignment paddle 57b can be adjusted by adjusting the acting positions of the first alignment paddle 57a and the second alignment paddle 57b. For instance, there is a method of setting the acting position of the first alignment paddle 57a closer to the processing tray 521 than the second alignment paddle 57b.

(85) In addition, the pressing forces of the first alignment paddle 57a and the second alignment paddle 57b can be adjusted by adjusting materials or thicknesses of the paddle parts 572 constituting the first alignment paddle 57a and the second alignment paddle 57b. For instance, there is a method of making the paddle part 572 of the first alignment paddle 57a using a material having a larger Young's modulus (longitudinal elastic modulus) than the material of the paddle part 572 of the second alignment paddle 57b. In addition, if the paddle parts 572 of the first alignment paddle 57a and the second alignment paddle 57b are made of the same material, there is a method of setting the thickness of the paddle part 572 of the first alignment paddle 57a larger than that of the paddle part 572 of the second alignment paddle 57a.

(86) Note that two first alignment paddles 57a are disposed here, but it may be possible to dispose one first alignment paddle 57a in the middle part in the sheet width direction. In addition, although one pair of the second alignment paddles 57b is disposed outside the first alignment paddles 57a, it may be possible to dispose two or more pairs of the second alignment paddles 57b.

(87) Although the embodiments of the present disclosure are described above, the scope of the present disclosure is not limited to these, but can be variously modified for implementation within the scope of the invention without deviating from the spirit thereof. For instance, in the first embodiment described above, the post-processing control unit 10 automatically adjusts the acting position of the alignment paddle 57 based on the output information of the sheet S, but it may be possible that the user can adjust the acting position of the alignment paddle 57 at any timing. For instance, there is a structure in which a conveying power adjusting mode is provided to the operation panel 202, in which the conveying power of the alignment paddle 57 is switched in three

steps of “small”, “middle”, and “large”, so that the user can select the mode in accordance with a situation of the stapling process (alignment property) in the sheet stapling unit **92**.

(88) In addition, in the embodiment described above, the case where the sheet stapling unit **92** performs the stapling process on the sheets S stacked on the processing tray **521** is exemplified, but the present disclosure is not limited to this, and may have a structure in which a shift discharge process or the folding process is performed on the sheets S stacked on the processing tray **521**.

(89) In addition, in the first embodiment described above, information about a type of the sheet S among the output information of the sheet S is input from the operation panel **202** of the image forming apparatus **200**, but it may be possible that the output information of the sheet S is automatically obtained. For instance, a media sensor is disposed at any position in the sheet conveying path from the image forming apparatus **200** to the sheet post-processing device **1**, and the output information of the sheet S such as a size, a thickness, a basis weight, a grain direction, surface smoothness, or amount of ink of the sheet S carried into the sheet post-processing device **1** from the image forming apparatus **200** can be detected using the media sensor.

(90) For instance, in order to detect the thickness of the sheet S, a laser coaxial displacement meter can be used as the media sensor, which detects the thickness by sandwiching the sheet S using two optical sensors.

(91) In order to detect the basis weight of the sheet S, a basis weight sensor can be used as the media sensor, which measures the basis weight based on transmittance of light through the sheet S. Note that depending of type of the sheet S, the relationship between the light transmittance and the basis weight varies, and hence it is necessary to select an optimal basis weight conversion formula for each type of the sheet S. In addition, by dividing the basis weight (g/m.sup.2) by the thickness (m), it is possible to calculate density (g/m.sup.3) of the sheet S.

(92) In order to detect the grain direction of the sheet S, an image sensing device is used as the media sensor, for example, and a transmission image is captured when LED light is emitted to the backside of the conveyed sheet S, and the captured transmission image is compared with a datum image stored in the main body control unit **203** in advance, so as to detect the grain direction of the sheet S. Alternatively, an ultrasonic measuring device is used as the media sensor, and a reflection waveform when an ultrasonic wave is emitted to the sheet S is measured and compared with a reference waveform stored in the main body control unit **203** in advance, so that the grain direction of the sheet S can be detected.

(93) In order to detect the surface smoothness of the sheet S, it is possible to use a surface property sensor that discriminates surface characteristics of the sheet S using optical reflection characteristics. In general, a sheet S such as plain paper or matte paper having low smoothness (rough surface) has reflection characteristics of perfect diffusion. In contrast, a sheet S such as glossy paper having high smoothness (high glossiness) has a state in which positive reflection and diffusion are mixed. The surface property sensor detects the surface property of the sheet S using the difference of the reflection characteristics described above.

(94) In order to detect type of the sheet S, a paper type discrimination algorithm is used, in which the type of the sheet S is discriminated using density, surface property (positive reflection light, diffusion light), basis weight, and the like of the sheet S.

(95) In addition, the amount of ink on the sheet S can be detected using a moisture meter as the media sensor, which detects water content in the sheet S.

(96) In addition, in the embodiment described above, the inkjet recording device is exemplified as the image forming apparatus **200**, but an electrophotographic printer or copier can also be used as the image forming apparatus **200**. Note that in the inkjet recording method in which ink is ejected to the sheet S, the conveying load of the sheet S tends to vary more than in the electrophotographic method. Therefore, the present disclosure is particularly useful for the sheet post-processing device **1** connected to the inkjet recording device as the image forming apparatus **200**.

(97) The present disclosure can be used for a sheet post-processing device that performs predetermined post-processing on a plurality of sheets.

Claims

1. A sheet post-processing device comprising: a conveying member configured to convey a sheet; a processing tray configured to stack a plurality of the sheets carried in by the conveying member in a predetermined carry-in direction; a processing section configured to perform predetermined post-processing on the sheets placed on the processing tray; a reference plate configured to abut an end edge of the sheet carried in onto the processing tray, on a downstream side in an alignment direction opposite to the carry-in direction, so as to align the sheet in the carry-in direction; an alignment member including a paddle holder disposed above the processing tray so as to be supported in a swingable manner in the carry-in direction, and an alignment paddle having a paddle main body supported in a rotatable manner at a swinging end of the paddle holder and one or more paddle parts protruding from the paddle main body in a radial direction, so as to move the sheet carried in onto the processing tray in the alignment direction for aiding alignment of the sheets; a holder drive unit configured to swing the paddle holder, so as to move the alignment paddle in a reciprocating manner between a reference position to retreat the paddle part above a top sheet placed on the processing tray and an acting position to allow the paddle part to contact with the top surface of the sheet; and a control unit configured to control the holder drive unit, wherein when the number of the sheets carried in onto the processing tray is increasing, the control unit continuously moves the acting position of the alignment paddle in a direction separating from the processing tray, the alignment paddle includes one or more first alignment paddles disposed in a middle part in the sheet width direction, and one or more pairs of second alignment paddles disposed outside the first alignment paddles in the sheet width direction, the first alignment paddle moves the sheet in the alignment direction while pressing a top surface of the sheet by a first pressing force, the second alignment paddle moves the sheet in the alignment direction while pressing the top surface of the sheet by a second pressing force, and the first pressing force is larger than the second pressing force.
2. The sheet post-processing device according to claim 1, wherein the control unit corrects the acting position of the alignment paddle based on output information about characteristics of the sheet.
3. The sheet post-processing device according to claim 2, wherein the output information about characteristics of the sheet includes at least one of a size, a thickness, a basis weight, a grain direction, and surface smoothness of the sheet.
4. The sheet post-processing device according to claim 1, wherein the processing section is a stapling unit that performs a stapling process to staple the plurality of sheets placed on the processing tray at a predetermined position with a staple needle.
5. The sheet post-processing device according to claim 1, wherein the acting position of the first alignment paddle is closer to the processing tray than the acting position of the second alignment paddle.
6. The sheet post-processing device according to claim 1, wherein the paddle part of the first alignment paddle is made of a material having a larger Young's modulus than the paddle part of the second alignment paddle.
7. The sheet post-processing device according to claim 1, wherein the paddle part of the first alignment paddle is made of the same material as the paddle part of the second alignment paddle, and has a larger thickness than the paddle part of the second alignment paddle.
8. An image forming system comprising: an image forming apparatus configured to form an image on a sheet; and the sheet post-processing device according to claim 1, configured to perform predetermined post-processing on the sheet after image formation by the image forming apparatus.

9. An image forming system comprising: an image forming apparatus configured to form an image on a sheet; and the sheet post-processing device according to claim 1, configured to perform predetermined post-processing on the sheet after image formation by the image forming apparatus, wherein the image forming system further comprises an input section for inputting output information about characteristics of the sheet.
10. The image forming system according to claim 9, wherein the acting position of the alignment paddle is correctable at any timing by an input operation to the input section.
11. An image forming system comprising: an image forming apparatus configured to form an image on a sheet; and the sheet post-processing device according to claim 2, configured to perform predetermined post-processing on the sheet after image formation by the image forming apparatus, wherein the image forming apparatus is an inkjet recording device configured to perform image formation by ejecting ink onto the sheet, and the output information about characteristics of the sheet includes amount of ink used for image formation on the sheet.
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